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Broadband RF Design with MEEP: A Set of 3D-Printed Open-Source RF Horns in the multi-GHz Bandwidth

Jordan C. Hanson [1]*, **Adam Wildanger**[1]

[1] Whittier College, Whittier, CA, USA (www.whittier.edu)
* Correspondence: jhanson2@whittier.edu

Abstract

Radio-frequency (RF) antenna design and fabrication traditionally proceeds in three phases. First, the design performance is modeled with expensive, proprietary software. Second, the design is built using precision metal machining or printed circuit board (PCB) fabrication. Third, the result is characterized using RF measurement tools. The process can include machine learning for optimization. We have developed an open-source alternative process that utilizes the MIT Electromagnetic Equation Propagator (MEEP) package for design and simulation, and open-source 3D printing with conductive polymer for fabrication. Using this process, we designed and fabricated a set of broadband RF horn antennas. The radiation patterns and the VSWR match our MEEP calculations, while the cross-polarization ratio matches theoretical expectations. These results indicate that our creations are linearly polarized, broadband RF horn antennas with bandwidth [5.5 - 18] GHz, or a fractional bandwidth of 106.4%. Because the maximum frequency of our network analyzer is 18 GHz, the absolute and fractional bandwidths are actually lower limits. Our MEEP calculations indicate the fractional bandwidth could be as high as 120%. Our results represent one of the first designs fabricated exclusively via conductive additive manufacturing, circumventing the need for electroplating processes. Future work will include the assembly of the set of RF horns into a phased array.

Keywords: Computational Electromagnetism (CEM), Additive Manufacturing, MEEP, RF Engineering, Open-Source Design

1. Introduction

Broadband RF antennas are ubiquitous tools within scientific instrumentation and communications. Traditionally, these are designed with expensive, proprietary software packages [1,2]. Designs are fabricated by cutting metal with precision machine tools (subtractive manufacturing). Compared to additive manufacturing techniques, subtractive manufacturing can be costly for rapid prototyping [3,4]. Open-source additive manufacturing boosts cost efficiency in both the design and fabrication, provided the 3D printing filament has sufficient conductivity.

Development of new RF antenna designs requires exploration of conductor shapes that set the boundary conditions for radiated signals. Developing new shapes that meet design requirements has been aided by machine learning (ML) strategies [5–7]. While mixing new ML algorithms into open CEM code is straightforward, incorporating them into proprietary software is often challenging. Ideally, RF engineers would keep the functionality of proprietary software, while retaining the flexibility to incorporate new ML techniques. Creating a design process based on open CEM tools achieves this goal.

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Previously, J. C. Hanson (2021) demonstrated that the open-source MEEP software package may be used as an RF antenna design tool [8]. MEEP operates via the FDTD algorithm for Maxwell's Equations on a Yee lattice [9]. The key to the technique is that MEEP is written to account for the scale-invariance of Maxwell's Equations. MEEP is normally used for designs with wavelengths at the μm scale. Designs cm wavelengths may be created by reinterpreting length units in MEEP. Using this approach, J. C. Hanson produced RF phased array designs in two and three dimensions using MEEP functionality [8]. The designs included broadband RF horn elements in the [0.5-5] GHz bandwidth. Further, J. C. Hanson demonstrated that open-source CAD software may be used to create complex designs that can be 3D printed [10].

In recent years, RF antenna systems have begun to be fabricated with 3D printing. The authors of [11] presented a hybrid strategy that combined traditional PCB fabrication, non-conductive 3D printing, and metallization of 3D printed structures. The assembled structure was a low-cost, electronically steerable MIMO array for mm-Wave 5G applications. The authors of [12] produced a waveguide-fed holographic beam-forming antenna using a dual print-head 3D printing system that combined non-conductive PLA supports with conductive filament. The result was a steerable beam at frequencies near 10 GHz. The authors of [13] produced a Mills-Cross, frequency-diverse, metasurface antenna capable of electronic steering in the K-band, using a mixture of non-conductive PLA and conductive filament. Finally, the authors of [14] printed a holgraphic metasurface antenna using the same technique found in [13] for Fresnel-zone focusing on and off-axis in the X-band.

In this work, we present the first open-source, broadband, 3D printed set of RF horns, designed with MEEP and fabricated entirely from conductive 3D-printer filament. Note that the authors are MEEP users, but not MEEP developers. Where appropriate, we have cited the creators of MEEP [9]. In Sec. 2, we explain the parametric design, CAD, and fabrication techniques. In Sec. 3, we show that the **E**-plane and **H**-plane radiation patterns, the VSWR, and the cross-polarization ratios match predictions. In Sec. 4, we discuss the limitations of our analysis, summarize key results, and discuss applications and future directions for this research.

2. Methods and Materials

The goal of this work was to use open CEM tools to design, model, and fabricate linearly polarized RF horn antennas in the [5-20] GHz range via additive manufacturing. Figure 1 contains a diagram of the design and fabrication process. Using parametric CAD design, we can specify antenna shapes using analytic functions that are then translated into the open CAD design tool kLayout. Antenna designs created with kLayout can be imported into MEEP via the GDSII file format. GDSII files can also be translated into 3D printer files, ensuring that the device and model match. In Fig. 2 (left), the CAD design in kLayout is shown. In Fig. 2 (middle)-(right), the model is rendered using Mayavi.

Our open fabrication technique hinges on the existence of commerically available conductive 3D printer filament. Multi3D[1] has shown that a copper-doped thermoplastic serves as a 3D printer filament with resistivities of $\approx 10^{-1}\ \Omega$ cm, depending on the printing pattern. Using a tungsten-tipped 3D printing head and Electrifi filament from Multi3D, we can complete antenna prints with volumes of ≈ 350 cm^3 in several hours. The printed structures are fitted with RF connectors and tested as receivers on custom mounts that swivel in zenith and azimuthal angles. A network analyzer with bandwidth [0-18] GHz was used to measure complex S11 and S21 parameters, with a calibrated log-periodic dipole array (LPDA) antenna as the transmitter.

[1] See https://www.multi3dllc.com

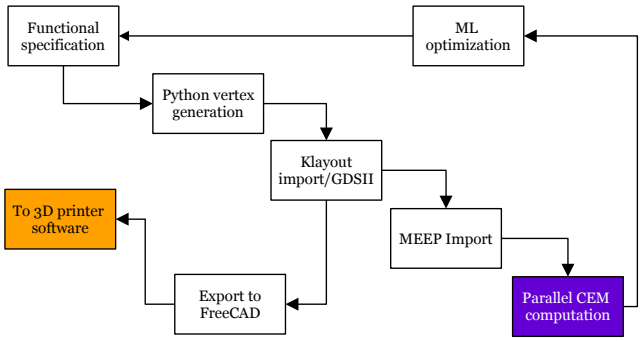


Figure 1. Flow diagram for the open-source design and fabrication process.

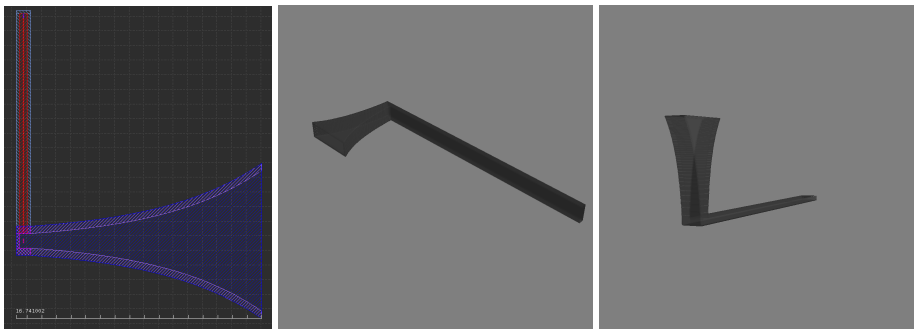


Figure 2. (Left) The kLayout CAD design for the RF horn (top view), with a coaxial cable attached. The x-direction is to the right, the y-direction is up, and the z-direction is out of the page. The white ruler gives 16.741 cm. (Middle)-(Right) Viewing the RF horn and cable model from alternate angles.

2.1. CEM Design and Open-Source CAD

The open-source CEM design begins with choosing parametric design parameters for the exponential RF horn and importing them into kLayout (Fig. 1-2). Let the origin refer to the center of the left edge of the cavity in Fig. 2 (left). The horn cavity is the rectangular volume where the coaxial cable attaches, with xy-dimensions $c_l \times c_w$ (Tab. 1). The horn surfaces are the curved structures that connect the cavity to open space, with overall length s_l in the x-direction, including the cavity. The opening is the area where the surfaces stop and radiation exits the horn. The opening has width w and height z in the y and z-directions, respectively. Using these variables, Equations 1 and 2 specify the shape of the RF horn surfaces in the xy-plane:

$$f(x) = \frac{c_w}{2} \exp(k(x - c_l)) \tag{1}$$

$$k = (s_l - c_l)^{-1} \ln\left(\frac{w}{c_w}\right) \tag{2}$$

The function $f(x)$ in Equation 1 describes the upper surface in Fig. 1, while $-f(x)$ describes the lower surface. Table 1 contains the values for c_l, c_w, s_l, and w. The RF horn is designed to be linearly polarized in the y-direction. The RF horn design has a height z, out of the page in Fig. 2 (left), and each conductive and dielectric surface, including the coaxial cable, has a specified thickness in the xy-plane and z-direction.

In previous studies [8], antenna structures were instantiated directly as native MEEP objects. In the present workflow, the antenna geometry is instead generated parametrically and transferred to kLayout, where it can be inspected and exported in GDSII format. The same geometry is then used for the MEEP calculation and for conversion to a printable 3D model. This approach preserves the flexibility of a scripted parametric design while

Table 1. Design parameters for the RF horn antenna.

Parameter	Variable Name	Value [cm]
Cavity Length	c_l	0.77
Cavity Width	c_w	1.00
Surface Length	s_l	16.52
Opening Width	w	9.59
Height	z	2.0

avoiding the need to reproduce a complex antenna from many simple MEEP primitives. The implementation code used to generate and simulate the antenna is maintained in the open-source project repository cited in the Data Availability Statement.

2.1.1. Radiation Pattern Calculations

We compute the antenna radiation pattern in MEEP using near-to-far field projection[2]. A `NearToFarRegion` encloses the radiating structure, and the `meep.get_farfield()` routine is used to evaluate the complex electric and magnetic fields in the far field. In the coordinate system of Fig. 2, the horn boresight is the $+x$ direction, the xy-plane is the **E**-plane, and the xz-plane is the **H**-plane.

The time-averaged Poynting vector is

$$\mathbf{S} = \frac{1}{2}\,\mathrm{Re}(\mathbf{E} \times \mathbf{H}^*), \tag{3}$$

and the radiated power used for the angular pattern is the component projected along the radial direction,

$$P_\mathrm{r} = \mathbf{S} \cdot \hat{\mathbf{r}}. \tag{4}$$

For the **E**-plane, $\hat{\mathbf{r}} = (\cos\phi, \sin\phi, 0)$, while for the **H**-plane, $\hat{\mathbf{r}} = (\cos\theta, 0, \sin\theta)$. The normalized radiation pattern is reported in decibels as

$$P_\mathrm{norm}(\alpha) = 10\log_{10}\left[\frac{P_\mathrm{r}(\alpha)}{\max(P_\mathrm{r})}\right]. \tag{5}$$

The corresponding source code is maintained in the open-source repository cited in the Data Availability Statement.

2.1.2. VSWR Calculations

The voltage standing wave ratio (VSWR) is expressed in terms of the magnitude of the complex reflection coefficient Γ at the antenna input port [17]:

$$\mathrm{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}. \tag{6}$$

In the MEEP model, a broadband excitation is launched through the modeled coaxial feed. Flux-monitor normalization is used to separate the incident and reflected contributions at the input line, from which Γ and the corresponding VSWR are calculated. The updated model is fully three-dimensional and uses the same imported CAD geometry as the radiation-pattern calculations. Implementation details and source code are provided in the open-source repository cited in the Data Availability Statement.

[2] See https://meep.readthedocs.io/en/master/Python_Tutorials/Near_to_Far_Field_Spectra

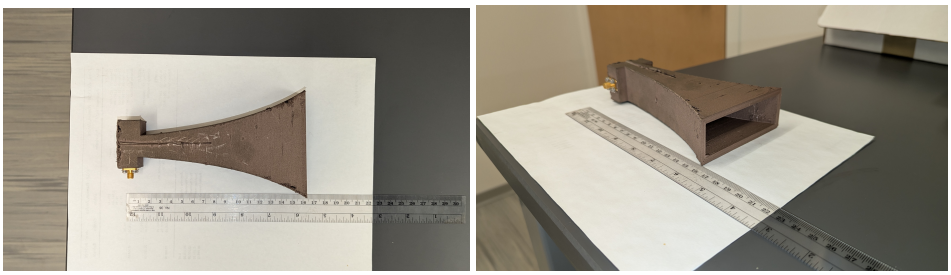


Figure 3. (Left) The Prusa XL 3D printer beginning a print with the Electrifi filament. (Middle) The finished RF horn with SMA connector. (Right) The finished RF horn viewed with opening visible.

2.2. Fabrication Technique

The parametric horn geometry was converted from the kLayout/GDSII design into a printable 3D model and fine-tuned for fabrication. The antennas were printed directly from Multi3D Electrifi conductive filament on a Prusa XL equipped with a tungsten-tipped nozzle. Electrifi was printed at approximately 135–145°C without bed heating. Because the horn is tall and narrow, masking tape with adhesive and removable buttresses were used to improve build-plate adhesion and mechanical stability during printing. After printing, the temporary supports were removed and an SMA connector was attached to the cavity so that the center conductor extended into the feed region. No secondary electroplating or conductive coating was applied to the radiating structure. One example of the RF horns is shown in Fig. 3, with a ruler for scale.

3. Results

Using the techniques described in Sec. 2.2, we fabricated eight RF horns with the parameters in Tab. 1. As listed in Tab. 1, the RF horn is 16.5 cm long (see ruler in Fig. 3). The dimensions of the opening are 2.5 x 8.5 cm, including the 0.25 cm thickness in the z-direction. The SMA connector is mounted to the cavity, with a coaxial pin extending into the chamber such that the radiation is linearly polarized in the y-direction (see Sec. 2.1).

3.1. Radiation Patterns

3.2. Cross-Polarization Ratios

The cross-polarization ratio was measured with the 3D printed RF horn and LPDA arranged in an S21 measurement configuration. The RF horn was the receiver, and bolted to the mount such that an inclinometer would measure cross-polarization angle. The prediction for the dependence of received power versus cross-polarization angle is known from optics and antenna theory. Let P_r be the received power, P_t be the transmitted power, and θ be the cross-polarization angle. These are related by

$$P_r = P_t \cos^2 \theta \tag{7}$$

When normalized, the ratio P_r/P_t in dB is equivalent to normalized S21. The results are shown in Fig. 5, and show good agreement with Eq. 7. The prediction from Eq. 7 tends to negative infinity in dB. In this regime we find a cross-polarization ratio approaching -15 dB, which is comparable to most side-lobes in broadband RF antennas. A good reference for Eq. 7 may be found in the section on polarization mismatch in [18].

4. Conclusions

This work establishes an open-source workflow coupling parametric antenna design, kLayout/GDSII geometry, MEEP electromagnetic modeling, and direct conductive additive manufacturing. The RF horn is printed directly from conductive Electrifi filament without

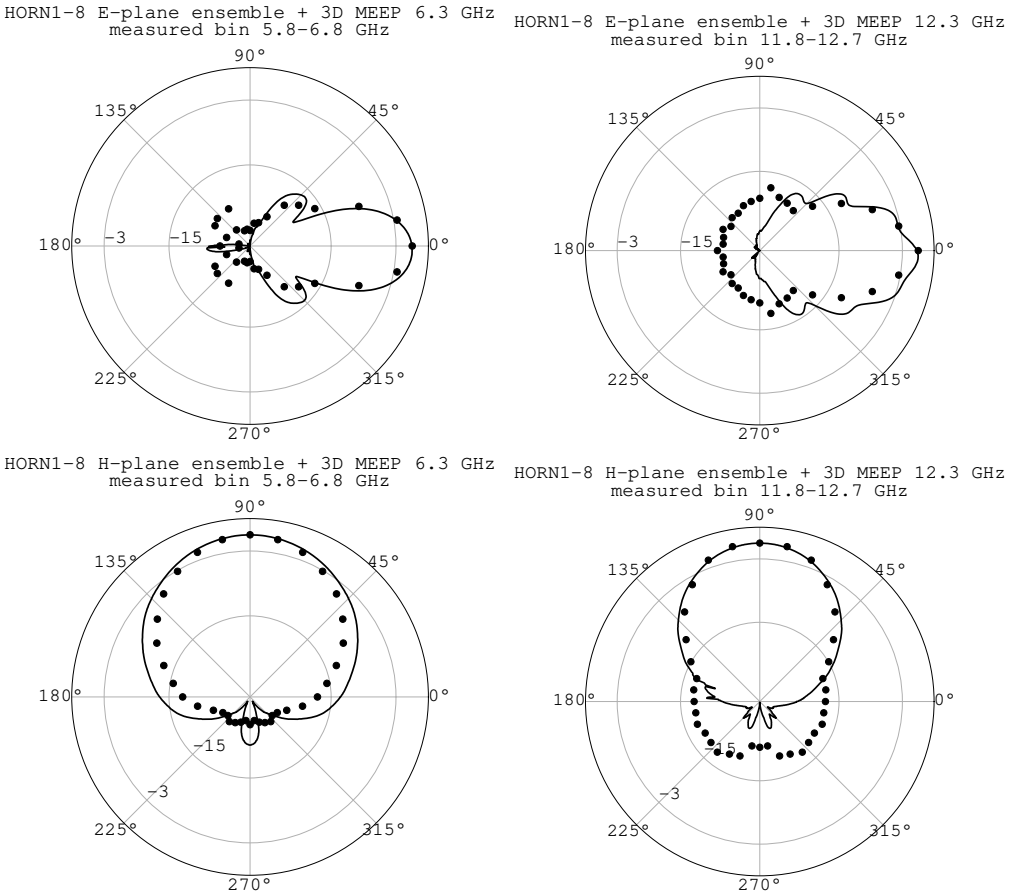


Figure 4. ss

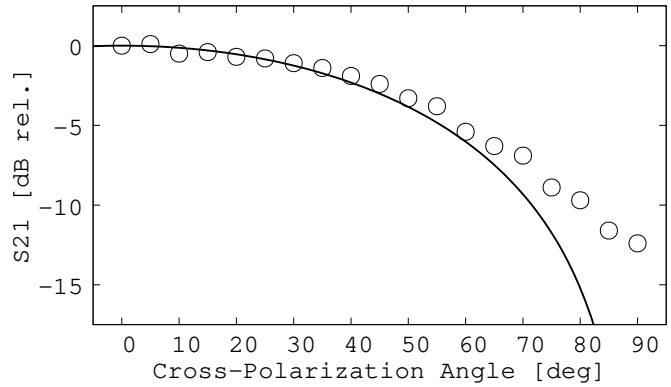


Figure 5. Circles: measured cross-polarization ratio in dB relative to maximum power. Line: Prediction from Equation 7.

a secondary metallization step, and the resulting hardware can be modeled using the same geometry used for fabrication. The retained cross-polarization measurement follows the expected $\cos^2\theta$ dependence for a linearly polarized antenna.

The broadband RF characterization and the updated comparison between the measured antennas and the three-dimensional MEEP model will be incorporated into the revised Results and Discussion using the expanded measurement set.

Author Contributions: J. C. Hanson (JCH) conceived of the open-source design process, produced the MEEP calculations that led to the RF horn design, identified the conductive 3D printer filament, incorporated open-source CAD, and produced fully three-dimensional CEM models to compute the VSWR and radiation patterns for the fabricated designs. A. Wildanger (AW) sourced materials, and imported CAD designs into 3D printing format, and completed 3D prints to produce the designs. Together, JCH and AW collected data using the RF measurement tools. JCH used the data to show that the MEEP calculations match the lab measurements. AW is exploring new models to be printed, including larger RF horn models.

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Data Availability Statement: Code and other resources available at https://github.com/918particle/Antenna-MEEP-project.

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

CEM	Computational Electromagnetism
HPC	High-Performance Computing
LPDA	Log-periodic Dipole Array
MEEP	MIT Electromagnetic Equation Propagator
NEEC	Naval Engineering Education Consortium
NSWC Corona	Naval Surface Warfare Center, Corona Division
RF	Radio-frequency
PLA	Polylactice Acid
PETG	Polyethylene Glycol
UAV	Unmanned aerial platforms
USD	United States Dollars

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